Sree Siddaganga College of Arts Science and Commerce, tumkur

Department of Computer Science

Project Synopsis

Project Title : AI-Based Phishing and Scam Detection System

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Project Title

AI-Based Phishing and Scam Detection System: Using Machine Learning and Natural Language Processing

1. Introduction

The rapid growth of digital communication, online banking, social media, and e-commerce has also led to an increase in phishing attacks and online scams. Cybercriminals commonly use fraudulent websites, emails, SMS messages, and social media messages to trick users into revealing sensitive information such as passwords, OTPs, banking details, and personal data. Many users find it difficult to identify such threats before interacting with them.

The proposed project, **AI-Based Phishing and Scam Detection System**, is a web-based application that uses **Machine Learning and Natural Language Processing (NLP)** to identify potentially harmful URLs and scam messages. The system analyzes the characteristics of a submitted URL or the content of a message and provides a risk classification along with an understandable explanation of the detected indicators.

2. Problem Statement

Phishing and online scams are becoming increasingly sophisticated and can closely resemble legitimate websites and messages. Traditional security mechanisms may not always be easily accessible or understandable to ordinary users. There is therefore a need for a simple and intelligent system that can assist users in identifying potentially malicious URLs and scam-related messages before they interact with them.

3. Objectives

The main objectives of the project are:

- To develop a web-based platform for detecting phishing URLs and scam messages.
- To extract relevant features from URLs for phishing classification.
- To apply Machine Learning algorithms to classify URLs as legitimate or potentially malicious.
- To use NLP techniques to analyze suspicious message content.
- To generate a risk score and explain the factors contributing to the prediction.
- To store users' scan results and history using a MySQL database.
- To compare different Machine Learning algorithms and evaluate their performance using suitable metrics.

4. Proposed System

The proposed system will provide users with a web interface where they can submit a URL or suspicious message for analysis.

For **URL detection**, the system will extract features such as URL length, number of special characters, domain characteristics, presence of an IP address, suspicious keywords, HTTPS usage, and other relevant attributes. These features will be provided to a trained Machine Learning model, which will classify the URL and generate a risk score.

For **scam message detection**, the system will preprocess the submitted text and apply NLP techniques to identify suspicious linguistic patterns, keywords, urgency-based language, fraudulent offers, requests for sensitive information, and other characteristics associated with scams.

The result will be presented in a simple format such as:

Classification: Potential Phishing
Risk Score: 91%
Risk Level: High

The system may also provide the reasons that contributed to the classification.

5. Technologies Used

Frontend

- HTML
- CSS

- JavaScript

Backend

- Python
- Flask

Artificial Intelligence

- Machine Learning
- Natural Language Processing

Machine Learning Libraries

- Pandas
- NumPy
- Scikit-learn

Database

- MySQL

Development Tools

- Visual Studio Code
- Git/GitHub

6. Major Modules

Module 1: Phishing URL Detection

- Dataset collection and preprocessing
- URL feature extraction
- Machine Learning model training
- URL classification
- Risk-score calculation
- Model evaluation

Module 2: Scam Message Detection

- Text dataset collection
- Text preprocessing
- NLP feature extraction

- Scam message classification
- Risk assessment

Module 3: Web Interface

- User registration and login
- URL analysis page
- Message analysis page
- Results page
- Dashboard
- Scan history

Module 4: Backend and Database

- Flask backend
- API development
- MySQL integration
- User management
- Scan-history management
- Integration of frontend and AI models

7. Methodology

The system will follow these general steps:

Data Collection → Data Preprocessing → Feature Extraction → Model Training → Model Evaluation → Backend Integration → Frontend Development → Database Integration → Testing

Different Machine Learning algorithms such as **Logistic Regression, Random Forest, and Support Vector Machine (SVM)** can be trained and compared. Evaluation will consider metrics such as **accuracy, precision, recall, F1-score, and confusion matrix** to determine the most suitable model.

8. Expected Outcome

The expected outcome is a functional web-based application capable of analyzing suspicious URLs and scam messages and providing users with an AI-based risk assessment.

The project will demonstrate the practical application of **Artificial Intelligence, Machine Learning, NLP, Python, web development, and database technologies** to a real-world cybersecurity problem.

9. Future Enhancements

The system can be extended in the future with:

- Screenshot-based scam detection using OCR
- Browser extension for real-time URL checking
- Email analysis
- Multilingual scam-message detection
- Real-time threat-intelligence integration
- Mobile application
- Improved deep-learning models
- Automatic reporting of confirmed malicious URLs

10. Conclusion

The **AI-Based Phishing and Scam Detection System** aims to provide users with an accessible tool for identifying potential online threats. By combining Machine Learning for URL analysis and NLP for scam-message detection, the system addresses a practical cybersecurity problem while providing significant scope for research, experimentation, and future development. The proposed technology stack of **Python, AI/ML, HTML, CSS, JavaScript, and MySQL** makes the system suitable for implementation as a four-member academic project.

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